A Progress Report

On

CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM

Phase-I — Progress Report – 1

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CERTIFICATE

This is to certify that the Phase-I progress report entitled "CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM" is being submitted by Divyansh Uniyal (2318766), Bhavya Gogia (2318662), Km Shivani (2318999), Abhishek Negi (2318178) for the partial fulfillment of the requirement for the award of the degree of B.Tech C.S.E. (Machine Learning & Artificial Intelligence) in Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun, and is a record of their own work carried out under my supervision. The matter embodied in the Phase-I progress report has not been submitted in any other university or institution for the award of any degree.

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CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the Phase-I progress report entitled "CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM" is being submitted for the partial fulfillment of the requirement for the award of the degree of B.Tech C.S.E. (Machine Learning & Artificial Intelligence) in Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun, is our original work and has not been submitted to any other university or institution for the award of any degree or diploma.

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CHAPTER 1

Introduction

1.1 Background of the Project

Urban road traffic in India is governed at its intersections. A signalised junction decides, several times a minute, which approach is allowed to discharge and for how long, and that decision determines the delay experienced by every vehicle upstream of it. The great majority of signalised junctions in Indian cities run a fixed-time plan: the cycle length and the green split are computed once from a manual turning-movement count, encoded into the controller, and then held for months or years. The method behind those plans is Webster's, published in 1958, and it remains the practical basis of signal timing in Indian design practice.

A fixed-time plan is correct on average and wrong most of the time. Demand at an urban junction varies across the day, across the week and with weather, events and incidents, while the plan does not. The visible consequence is green time held on an empty approach while a saturated approach waits, and it is not a failure of the arithmetic — it is the cost of a plan that cannot see the junction it is controlling.

The traffic the plan must serve is also not the traffic the method assumes. Indian urban streams are heterogeneous and non-lane-disciplined. A single approach carries two-wheelers, auto-rickshaws, e-rickshaws, cars, light commercial vehicles, buses, trucks and, on many roads, pedestrians and animals in the carriageway. These modes occupy different road space and discharge at different rates, and they do not queue in the lanes that are painted on the road. Indian practice already recognises this: the Indian Highway Capacity Manual publishes passenger-car-unit equivalences precisely because counting vehicles is not the same as measuring how much road is occupied.

What has changed, and what makes this project possible, is sensing. Surveillance cameras are already installed at a large and growing number of Indian junctions for enforcement and monitoring. They are a sensor the signal does not currently use. Inductive loops, radar and magnetic probes — the instrumentation that the mainstream traffic-forecasting literature assumes — require trenching, calibration and maintenance budgets that most of these junctions do not have. A camera that is already mounted, already powered and already pointed at the stop line is the one sensor that can be used without installing anything.

1.2 Problem Statement

Given only the video stream of a single fixed camera already installed at a signalised urban intersection, predict the congestion level of each approach lane sixty seconds into the future, accurately enough for a signal controller to act on the prediction rather than on a measurement of the present — and do so under heterogeneous, non-lane-disciplined traffic, on hardware a municipal body could afford, without transmitting or storing a single raw video frame.

Each clause of that statement is a constraint rather than a preference. A single camera, because a method that needs several cameras per junction cannot be deployed at the junctions this targets. Per-lane output, because a controller allocates green to an approach and a scene-level congestion score gives it nothing to act on. Sixty seconds, because a signal decision takes effect over the next phase and a forecast of the present is a measurement, not a forecast. No raw frames leaving the device, because a system that moves footage of a public road off the camera is a surveillance system regardless of what it is used for.

1.3 Motivation

The cost of the current arrangement is paid in delay, fuel and emissions at every junction, every day, and it is paid most heavily by the modes that are least able to absorb it. A two-wheeler or an auto-rickshaw waiting out a green phase allocated to an empty approach is not an abstraction; it is the ordinary experience of an Indian commute.

The reason this has not already been solved is not that adaptive control is unknown. It is that adaptive control has been built for a road environment that is instrumented, lane-disciplined and homogeneous, and the published methods inherit those assumptions through their inputs. A controller that expects loop-detector occupancy cannot be installed where there are no loops. A forecaster trained on lane-disciplined motorway flow does not transfer to a junction where vehicles straddle the markings. The gap is not in the control theory; it is between the sensing the methods assume and the sensing that exists.

The second motivation is methodological, and it is why this is a research project and not an integration exercise. The architecture this project proposes fuses a convolutional and a transformer view of the same frame through a learned gate. Whether that gate learns anything interpretable — whether it shifts toward one branch as the scene becomes dense — is a question that can be answered with evidence, and answering it is worth more than another percentage point on a benchmark.

1.4 Existing Scenario

Four approaches are in use or in the literature, and each addresses part of the problem.

Fixed-time control, computed by Webster's method from a periodic manual count, is the deployed baseline at the junctions this project targets. Vehicle-actuated control, where a detector extends a green phase while vehicles are still arriving, is deployed where detectors exist. Sensor-based forecasting — the spatio-temporal graph literature — predicts flow or speed across a road network from loop-detector and probe time series. Vision-based congestion estimation classifies congestion directly from camera imagery, and is an active field rather than an opening.

The limitation each approach carries is stated against it below, because a limitation recorded apart from the approach it belongs to is easy to agree with and impossible to act on.

Fixed-time control. The plan cannot respond to the demand it is serving. Re-timing requires a fresh manual survey, so plans age; and the method's saturation-flow parameter is itself uncertain under heterogeneous traffic, where published values for a single intersection differ by a factor of two depending only on which passenger-car-unit convention is used.

Vehicle-actuated control. It requires the detector infrastructure that is absent at the target junctions, and it reacts to arrivals that have already happened rather than anticipating the next phase.

Sensor-based graph forecasting. It is accurate and well developed, and it assumes an instrumented network with a known road graph. Its inputs do not exist at an un-instrumented Indian junction, so its accuracy is not available there at any price.

Vision-based congestion estimation. It uses the sensor that does exist, and most published work classifies the scene as a whole, estimates present conditions rather than forecasting, and is evaluated on lane-disciplined traffic. None of those three gives a signal controller a per-lane forecast it can act on.

Reinforcement-learning signal control. It is a mature and heavily benchmarked literature, and its state vectors are built from instantaneous measurements. Whether an anticipatory input helps a controller, and how good that anticipation must be before it helps, is largely untested.

1.5 Need for the Proposed Work

The four approaches above do not compose into a solution, because the accurate ones need sensors that are absent and the deployable one does not produce what a controller needs. What is missing is a method that takes the sensor that is already installed and produces the quantity a controller can act on: a per-lane congestion level, one phase ahead, under traffic that does not respect lanes.

That is the gap this project is built for, and the engineering it implies is specific. Congestion must be defined on occupancy weighted by passenger-car units rather than on a raw vehicle count, because a count that treats a bus and a two-wheeler alike measures the wrong thing. The thresholds separating congestion levels must be calibrated per camera, because a threshold that is correct for one viewpoint is arbitrary at another. Evaluation must hold out whole cameras rather than shuffling windows, because a model that has seen a camera in training will report recognition as generalisation. And the detector the pipeline depends on must be fine-tuned for the vehicle classes Indian streams actually contain, because an off-the-shelf detector has never been trained to see an auto-rickshaw.

1.6 Project Objectives

The project has six objectives. Each is stated as something that will be built or established, and each is measurable in the sense that its absence would be visible.

O1. Assemble a fixed-camera corpus of heterogeneous traffic with per-lane congestion labels. Curate footage from permissively-licensed public sources, place lane regions by hand for each camera, derive per-lane passenger-car-unit weighted occupancy from detector counts, and label each lane's congestion level sixty seconds ahead. Splits are cut by source camera so that no camera appears in both training and test.

O2. Fine-tune a vehicle detector for the classes Indian streams contain. Train a single-stage detector on an eight-class Indian taxonomy including auto-rickshaw and e-rickshaw, and report per-class accuracy with support counts rather than a single averaged figure.

O3. Design and train a multimodal spatio-temporal forecaster. Combine a convolutional and a self-supervised transformer view of each frame through bidirectional cross-attention and a learned gate, model the sequence with a bidirectional recurrent encoder, and predict a congestion class per lane at a sixty-second horizon.

O4. Establish whether each component earns its place. Run an ablation in which every module can be disabled by configuration alone, including a linear-probe floor, and compare against count-based forecasters that consume only detector counts. Report the comparison whichever way it goes.

O5. Couple the forecast to a signal controller and measure the coupling. Train a reinforcement-learning controller on a simulated heterogeneous intersection with the forecast as part of its state, and measure against the best configuration of the fixed-time baseline rather than an arbitrary one.

O6. Keep the whole pipeline reproducible and privacy-preserving. Fix seeds across every library, commit raw per-run results rather than summary tables, generate every reported table from those results by a committed script, and ensure no raw video frame is ever written to disk or transmitted.

1.7 Scope of the Project

1.7.1 Within scope

- Per-lane congestion forecasting at a sixty-second horizon from a single fixed camera, for a four-approach signalised intersection.
- A curated fixed-camera corpus of heterogeneous traffic, with per-lane labels derived from passenger-car-unit weighted occupancy.
- A vehicle detector fine-tuned for an eight-class Indian taxonomy.
- An ablation over the architecture, and a comparison against count-based forecasters that use no appearance information at all.
- A reinforcement-learning signal controller evaluated in simulation, with the forecast as an input to its state.
- An edge-to-server prototype in which only derived counts and predictions cross the network.

1.7.2 Outside scope

Stated explicitly, because an unstated exclusion reads as an unmet expectation at evaluation time.

- Field deployment on a live signal. Nothing in this project actuates a real traffic light, and no wait-time reduction is claimed for the real world.

- Multi-intersection coordination. The controller is single-junction; network-level cooperation is established work and is not attempted.
- Vehicle re-identification, tracking across cameras, number-plate recognition, or any identification of individuals or vehicles.
- Trajectory prediction for individual vehicles. The output is a per-lane congestion class, not a path.
- A production data platform. Storage is a single embedded database and the dashboard is a diagnostic surface, not an operations product.

1.8 Proposed Approach

The approach is sketched here in outline; the detailed design belongs to the design documents that follow this report. Seven stages, each producing an artefact the next one consumes.

Stage 1 — Corpus. Triage candidate footage, place per-camera lane regions by hand, detect and count vehicles per lane per step, convert counts to passenger-car-unit weighted occupancy, and label each lane's congestion level at the forecast horizon. Thresholds are calibrated per camera at the 33rd and 67th percentiles of that camera's own occupancy distribution.

Stage 2 — Detection. Fine-tune a single-stage detector on an eight-class Indian taxonomy, bootstrapping from a public Indian dataset so that corpus construction is not blocked on collecting new footage.

Stage 3 — Representation. Extract two views of each frame with frozen backbones — one convolutional, one self-supervised transformer — and cache the features, so that an ablation over the layers above them does not re-run the backbones.

Stage 4 — Fusion. Let each view attend to the other in both directions, and combine the two results with a learned gate whose value is logged and analysed rather than hidden inside the network.

Stage 5 — Temporal model and head. Encode the fused sequence with a bidirectional recurrent network and temporal self-attention, pool by attention rather than by taking the last state, and pool per-lane regions separately so the four lanes can receive four different predictions.

Stage 6 — Control. Train a policy-gradient controller on a simulated heterogeneous four-way intersection whose state includes the per-lane forecast, and compare it against the best-tuned fixed-time plan.

Stage 7 — Prototype. Run detection on an edge device, publish only derived counts over a message broker to a server that holds the forecaster and the controller, and fall back to a locally-embedded fixed-time plan if the link drops.

The order is a constraint and not a convenience. A standard cross-attention model is brought to convergence before the gate replaces it, and the gate before temporal attention is added, because the known failure mode of a project shaped like this one is an elaborate architecture with no experiments behind it. Every module is therefore disableable by configuration alone, which is what makes the ablation possible without editing code between runs.

1.9 Expected Outcome

What will exist at the end of the project, and what will be known. No figure is quoted: this report is written before the measurements, and a target stated here as though it were a result would be exactly the overclaim the project's own reporting rules forbid.

A labelled fixed-camera corpus. Per-lane congestion labels at a sixty-second horizon, with the threshold calibration recorded per camera and splits cut so that no camera is shared between training and test.

A detector tuned for Indian vehicle classes. Per-class accuracy reported with support counts, so that a strong average over common classes cannot hide a failure on the rare ones the taxonomy exists to add.

An answer on whether visual fusion earns its cost. The ablation and the count-only baselines answer this directly. A finding that appearance does not beat counts in this regime is a reportable result and will be reported as one.

Evidence about what the gate learns. The gate value is logged and analysed against scene density. A gate that collapses or tracks nothing is an informative negative result.

A controller result with its uncertainty. The control comparison is reported as a mean with a bootstrap confidence interval over repeated seeds against the best-tuned fixed-time baseline, with a paired test and an effect size, rather than as a single headline figure.

A reproducible package. Fixed seeds, committed raw per-run results, and every reported table generated from those results by a committed script.

CHAPTER 2

Literature Survey

The survey is organised by what each strand of work settles rather than by publication date, because the question this report has to answer is not what exists but what is left. Fourteen subsections follow: the task-level prior art first, then the forecasting paradigm this project departs from, then density estimation and detection, then the representation and fusion mechanisms it assembles — every one of which is published and is cited as such — then the control literature, the data situation and what can be done about it, how transfer and imbalance are measured, and the signal-timing ground the baseline stands on.

Each citation in this report points at a record that was retrieved and checked rather than recalled. The reference list is generated from a committed file of verified bibliographic records, and a citation whose record is missing fails the build rather than printing a number that points at nothing.

2.1 Congestion estimation from camera imagery

The task-level prior art is a camera and a congestion label, and it is an active field rather than an opening. Chakraborty et al. [1] classify congestion directly from traffic-camera still images with deep convolutional networks, establishing that the signal is present in a single frame from a fixed viewpoint. Kumar and Raubal [2] survey the use of deep learning across congestion detection, prediction and alleviation; the survey's existence is itself the finding that matters here, because it means a project claiming camera-based congestion estimation as novel would be claiming something the field settled years ago.

The closest work at the task level is Feng et al. [3], which assembles detection, tracking and prediction into one end-to-end pipeline driven by fixed low-resolution surveillance cameras. It shows the pipeline is buildable from the sensor this project assumes. What it does not do — and what none of this strand does — is produce a per-lane forecast under non-lane-disciplined traffic and hand it to a controller. The scene-level output and the lane-disciplined evaluation setting are where the opening is.

2.2 Spatio-temporal forecasting from sensor networks

The dominant traffic-forecasting paradigm treats the road network as a graph and the sensors on it as a multivariate time series. Yu et al. [5] apply graph convolution jointly over the spatial graph and the temporal axis; Li et al. [4] model propagation as a diffusion

process on a directed graph with a recurrent sequence-to-sequence decoder; Wu et al. [7] learn the adjacency matrix rather than taking it from the road geometry, removing the need for a correct hand-built graph; and Guo et al. [6] add attention over both axes so the model can weight periods and locations differently.

This strand is mature, well benchmarked, and consistently strong on its own datasets. Its relevance here is in what it assumes rather than what it achieves. Every method above consumes loop-detector or probe-vehicle readings from an instrumented network whose topology is known. At an Indian junction with a camera and no loops, those inputs do not exist, so the accuracy these methods report is not available at any price. That is the assumption this project departs from, and stating it that way is more defensible than claiming to outperform methods evaluated on different inputs.

2.3 Density estimation, and counting what cannot be separated

A strand the first version of this survey missed, and the nearest neighbour to what this project actually measures. Congestion here is passenger-car-unit weighted occupancy — a density, not a count — and estimating density directly from an image is a solved problem in another domain. Zhang et al. [8] regress a continuous density map with a multi-column network at several receptive-field scales, so that objects at different apparent sizes are handled by different columns. Li et al. [9] replace the multi-column design with dilated convolutions behind a truncated classification backbone, and report that density can be recovered where individual objects are no longer separable at all.

That last property is the one that matters at a saturated Indian approach. When vehicles overlap to the point where a detector cannot put a box on each, a counting pipeline degrades and a density estimator does not. This project counts rather than regresses density, because a count is what converts to passenger-car units and a density map does not carry vehicle class. That is a deliberate trade, recorded here as one: the detector's behaviour under saturation is a known risk, and this literature is where the alternative lives if that risk materialises.

2.4 Object detection as the pipeline's first stage

Every count in this project comes from a detector, so detection accuracy bounds everything downstream. Redmon et al. [10] reframe detection as a single regression over a grid, which is what makes per-frame inference on a live stream affordable; the single-stage family this project uses descends directly from it. Lin et al. [11] identify why single-stage detectors

had lagged — the loss is dominated by easy background — and show that reshaping it to down-weight easy negatives closes most of the gap.

Both results bear on a measured property of this project's detector rather than on a design choice. Accuracy on small, crowded objects is the known weakness of the single-stage family, and an obliquely-viewed junction at saturation is precisely that case. The imbalance Lin et al. correct is between object and background; the imbalance this project carries is between common and rare vehicle classes, which [43] addresses and which Section 2.12 returns to.

2.5 Frozen visual representations

Two families of frozen feature extractor are relevant. Residual convolutional networks [12] remain a strong local-texture encoder and are cheap enough to run per frame. Vision transformers [13] treat an image as a sequence of patches and model long-range relations directly, which is the property a congested scene needs: occupancy is a relation between distant parts of the frame, not a local texture.

Two later results shape the choice. Oquab et al. [15] show that self-supervised pre-training yields transformer features that transfer without labels, which matters when the backbone will be frozen and the downstream corpus is small — a frozen backbone makes representation quality the whole ceiling. Against that, Liu et al. [14] show a modernised pure convolutional network matching transformer performance, which is the adversarial citation this project must answer: if the transformer branch cannot be shown to contribute, it should not be in the model. That is precisely what an ablation with a convolution-only configuration is for.

2.6 Hybrid convolutional-transformer architectures

Combining the two families is established work, and the nearest neighbour is explicit about it. Peng et al. [19] run parallel convolutional and transformer branches and exchange information between them through a coupling unit at every stage. Chen et al. [16] fuse two transformer branches operating at different token scales using cross-attention on their classification tokens, establishing cross-attention as a dual-branch fusion mechanism. Dai et al. [17] stack convolution and attention within one network, and Mehta and Rastegari [18] interleave them for efficiency on mobile hardware.

The consequence for this project is a disclaimer that has to be made rather than avoided: a dual convolutional-transformer design fused by cross-attention is not a contribution in

2026, and a report that presented it as one would be contradicted by the first reviewer who knows this strand. What the architecture can contribute is evidence — an ablation that isolates whether the fusion does anything at all on this task, which these works do not answer for per-lane congestion forecasting.

2.7 Cross-attention and gated fusion

The mechanisms this project assembles are each published. Scaled dot-product attention and the multi-head transformer block [20] are the substrate. Lu et al. [21] introduce co-attentional layers in which each stream's queries attend to the other stream's keys and values, in both directions — which is exactly the bidirectional cross-attention this project uses, published in 2019 for vision and language. Tan and Bansal [22] build a cross-modality encoder on the same principle.

Gating the result is also established. Alayrac et al. [23] place learned gates on cross-attention layers to control how much cross-modal information flows into the main stream. The gate in this project is a convex combination of the two cross-attended streams, and as a mechanism it is standard.

What is not standard is the use made of it. In the works above a gate is an optimisation device, reported only through its effect on accuracy. Here the gate's value is treated as an output: it is logged per window, analysed against scene density, and reported whether or not it behaves as predicted. That reframing — mechanism as measurement — is the defensible contribution, and it survives the fact that every component of the mechanism is prior art.

2.8 Sequence models and the baselines that must be beaten

The temporal half of the problem has a long-settled default. Long short-term memory [24] remains the standard recurrent encoder for a sixty-step sequence of this length, and a bidirectional pair of such layers is the project's temporal encoder.

The more important citation in this strand is a baseline rather than a component. Gradient-boosted decision trees [25] are extremely strong on small tabular problems, and a per-lane vehicle count sequence is a small tabular problem. If a boosted-tree model reading detector counts matches a dual-backbone attention architecture reading pixels, the complexity of the latter is not justified, and the honest course is to report that. This is why count-only baselines are built in parallel with the model rather than afterwards.

2.9 Reinforcement learning for signal control

Learned signal control is a mature literature. Wei et al. [27] apply deep reinforcement learning to signal control on real-world data; Wei et al. [28] shape the reward with max-pressure theory so the learned policy inherits a known stability argument; Wei et al. [29] coordinate many intersections through graph attention; Zheng et al. [30] exploit the phase-competition symmetry so a policy generalises across intersection geometries; and Chen et al. [31] scale decentralised pressure-based control to city size. Wei et al. [32] survey the field and its evaluation practices.

The method this project uses for the controller itself is proximal policy optimisation [26], a standard on-policy algorithm, applied through a standard implementation. None of this is a contribution, and the report says so: single-intersection control with a policy-gradient method on a microscopic simulator is a well-solved problem.

The open question this strand leaves is about the state, not the algorithm. The works above build state vectors from instantaneous measurements — queue lengths, pressures, occupancies of the present. Whether a controller benefits from an *anticipatory* input, and how accurate that anticipation must be before the benefit appears, is largely untested. That question is answerable with a sensitivity curve over forecast quality, and it is the project's contribution on the control side.

2.10 Datasets, and why viewpoint is the real gap

The data situation is specific and it is the project's hardest constraint. Varma et al. [34] provide the reference Indian driving dataset, with the heterogeneous classes this project needs — including auto-rickshaw and animals — but captured from an ego-vehicle dashcam, because its agenda is autonomous navigation. A moving viewpoint cannot support per-lane regions that stay valid across a sixty-step window.

Conversely, the fixed-camera benchmarks have the right viewpoint and the wrong traffic. Wen et al. [35] provide a fixed elevated multi-object detection and tracking benchmark on Chinese traffic, and Tang et al. [33] a city-scale fixed multi-camera benchmark on United States traffic. Both are lane-disciplined and neither contains the vehicle modes an Indian stream does.

The intersection of the two properties — heterogeneous Indian traffic *and* a fixed elevated viewpoint, as continuous video rather than still frames — is genuinely under-

served, and that is a dataset gap rather than a modelling one. It is also the reason this project curates its corpus from licensed public sources instead of assuming a benchmark exists.

2.11 Synthetic data, and how far a foreign domain transfers

When the data a project needs does not exist, the literature offers two answers, and they are not the same answer. The first is to generate it. Richter et al. [38] extract pixel-accurate labels from a commercial game engine at no annotation cost; Ros et al. [39] build a synthetic urban corpus for the same purpose, and Gaidon et al. [36] clone real sequences into a virtual world so that the same scene exists in both. Dosovitskiy et al. [40] provide an open urban driving simulator with sensors and a control interface attached. Tobin et al. [42] show that deliberately randomising a simulator's appearance, rather than making it realistic, is often what makes a model transfer.

The second answer is to adapt rather than generate. Ganin et al. [37] train a feature extractor adversarially so that a classifier cannot tell which domain a sample came from, and Hoffman et al. [41] translate images between domains while enforcing cycle consistency and semantic preservation.

The finding common to both answers is that the synthetic-to-real gap is real and must be measured rather than assumed away: synthetic data substitutes for a large fraction of real annotation, and a small amount of real in-domain data recovers most of what remains. For this project the implication is specific, and it points away from generating frames. The deployment viewpoint and the deployment country are both available in real, openly-licensed data, so synthetic imagery would substitute for something already held. Where generation has no substitute is in producing a *controlled* variable — traffic at a chosen degree of lane discipline, which no observational dataset can contain, because a real road has whatever discipline it has and offers no counterfactual.

2.12 How transfer and imbalance are measured

Two of this project's methodological commitments are stated in Section 1.5 as though they were self-evident, and this literature is what makes them defensible. Holding out whole cameras rather than shuffling windows is a leave-one-domain-out protocol, and Gulrajani and Lopez-Paz [44] show why the protocol matters more than the method: under a careful and consistent evaluation, no domain-generalisation algorithm reliably beat plain empirical risk minimisation, and much of the reported progress came from differences in model selection rather than from the methods themselves. Zhou et al. [45] survey the field and its evaluation practice. The lesson taken here is not that generalisation is hopeless, but that the

split and the model-selection rule have to be fixed before any number is produced — which is why both are declared in advance.

The second commitment is reporting per-class accuracy with support counts rather than a single average. Cui et al. [43] re-weight a loss by the effective number of samples per class instead of by raw frequency, and show the advantage grows with imbalance. The limit of that method is the condition this project is actually in: a weighting scheme assumes every class is present to be weighted, and it cannot help a class that has no test instances at all.

2.13 Signal timing under heterogeneous traffic

The baseline this project must beat has to be constructed carefully, because its central parameter is uncertain. Webster's method [49] needs a saturation-flow value, and under non-lane-disciplined heterogeneous traffic that value is not a constant. Sugiarto et al. [46] measure base saturation flow on non-lane-based urban traffic and fit it to approach width, reporting passenger-car equivalents of 0.24 for a motorcycle and 0.78 for an auto-rickshaw, with start and end lost times of four to five seconds and about three seconds. Indian practice itself supplies two different conventions for the same quantity: the Indian Roads Congress geometric guidance [50] and the Indian Highway Capacity Manual [51] do not agree, and published measurements at a single intersection differ by roughly a factor of two depending on which is used.

The methodological consequence is that the fixed-time baseline cannot be built from one chosen saturation-flow figure. The published range has to be swept and the *best* configuration of the baseline taken as the thing to beat, because beating a badly-tuned baseline measures the tuning rather than the method.

2.14 Simulation and implementation substrate

Control cannot be evaluated on a live signal, so it is evaluated in microscopic simulation. SUMO [47] supports sub-lane lateral movement, which is what allows a simulated stream to be non-lane-disciplined rather than a homogeneous queue, and the residual gap between the simulated stream and a real Indian one is declared rather than ignored. The controller is implemented with Stable-Baselines3 [48], a reference implementation whose behaviour is documented and reproducible, and the detector with a current single-stage detection framework [52]. Reinforcement learning terminology throughout follows the standard treatment [53].

CHAPTER 3

Literature Survey Table

Table 3.1 compares the 30 works that bear most directly on this project. The author, year and title columns are generated from the verified bibliographic record, so a wrong citation cannot be introduced by editing the table. The methodology, setup, findings and limitation columns state each work's method and the assumption it makes — which is what determines whether its result is available at an un-instrumented Indian junction — rather than restating its reported numbers out of their experimental context.

Four further works cited in Section 2 are deliberately absent from the table: a mobile hybrid backbone and a second cross-modality encoder, each a near-duplicate of a row already present, and the simulator and reinforcement-learning library, which are tools rather than research results. All four remain in the references.

The table is set on landscape pages so that seven analytical columns are legible. The page margins are unchanged.

Table 3.1 Comparative analysis of 30 works related to the project

S. No.	Author(s) & Year	Research Paper Title	Methodology / Technique Used	Dataset / Experimental Setup	Key Findings	Limitations / Research Gap
1	P. Chakraborty et al., 2018	Traffic Congestion Detection from Camera Images using Deep Convolution Neural Networks	Deep convolutional classifiers (and a comparison against shallow features) applied to single still images from traffic cameras	Still images from a deployed highway traffic-camera network, labelled congested or uncongested	Congestion is recoverable from one frame of a fixed camera, without tracking or a calibrated scene model	Scene-level binary label, present-tense detection rather than a forecast, and no per-lane output a controller could act on
2	N. Kumar, and M. Raubal, 2021	Applications of deep learning in congestion detection, prediction and alleviation: A survey	Survey of deep-learning methods across congestion detection, prediction and alleviation, organised by task and input modality	Survey of published work; no single experimental setup	The field is large and active, and camera-based congestion estimation is established rather than open	A survey fixes the state of the art but answers no question; it is cited to bound the novelty claim, not to borrow a method
3	Y. Feng et al., 2024	Predicting spatio-temporal traffic flow: a comprehensive end-to-end approach from surveillance cameras	End-to-end pipeline chaining vehicle detection, multi-object tracking and short-horizon flow prediction	Fixed low-resolution urban surveillance camera video	Detection, tracking and prediction compose into one deployable pipeline driven only by an existing camera	Link- or scene-level flow rather than per-lane congestion, lane- disciplined setting, and the forecast is not consumed by a controller
4	B. Yu, H. Yin, and Z. Zhu, 2018	Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting	Graph convolution over the sensor graph combined with gated temporal convolution, in a fully convolutional spatio-temporal block	Loop-detector speed and flow time series on instrumented road networks	Spatial structure and temporal dynamics can be modelled jointly without recurrence, at low cost	Requires an instrumented network and a known road graph; neither exists at the target junctions, and no visual input is used
5	Y. Li et al., 2017	Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting	Traffic as a diffusion process on a directed graph, with diffusion convolution inside a recurrent encoder-decoder trained by scheduled sampling	Loop-detector speed series from instrumented freeway and arterial networks	Directed, asymmetric spatial propagation matters, and multi-step forecasting benefits from modelling it explicitly	Same infrastructure assumption; the directed graph must be supplied, and the method has no access to appearance
6	Z. Wu et al., 2019	Graph WaveNet for Deep Spatial-Temporal Graph Modeling	Learned adaptive adjacency matrix combined with dilated causal convolution over the temporal axis	Loop-detector series on instrumented networks	The spatial graph can be learned from data rather than taken from road geometry, removing the hand-built-graph requirement	Removes the need for a *correct* graph but not for the sensors; still sensor-only and lane-disciplined

S. No.	Author(s) & Year	Research Paper Title	Methodology / Technique Used	Dataset / Experimental Setup	Key Findings	Limitations / Research Gap
7	S. Guo et al., 2019	Attention Based Spatial-Temporal Graph Convolutional Networks for Traffic Flow Forecasting	Spatial and temporal attention over a graph convolutional backbone, with separate components for recent, daily and weekly periodicity	Loop-detector flow series on instrumented networks	Attention over both axes lets the model weight locations and periods differently, improving multi-horizon accuracy	Sensor-dependent; the periodicity components presume stable recurring demand patterns and a long observation history
8	K. He et al., 2016	Deep Residual Learning for Image Recognition	Residual connections allowing very deep convolutional networks to be trained by learning residual functions	Large-scale image classification and detection benchmarks	Depth becomes trainable, and the resulting features transfer broadly as a general-purpose local encoder	Convolutional inductive bias is local; relations between distant parts of a frame are reached only through depth
9	A. Dosovitskiy et al., 2020	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	Image as a sequence of flattened patches fed to a standard transformer encoder, with no convolutional prior	Large-scale supervised image classification pre-training, transferred to smaller benchmarks	Global relations can be modelled directly, and attention-based vision matches convolutional networks given enough pre-training data	Data-hungry without a convolutional prior; patch count fixes the token budget and therefore the spatial granularity
10	M. Oquab et al., 2023	DINOv2: Learning Robust Visual Features without Supervision	Self-supervised pre-training of vision transformers by self- distillation, producing features used without fine-tuning	Large curated unlabelled image corpus; evaluated by frozen-feature transfer	Frozen self-supervised features transfer strongly without any labels, which is what makes a frozen backbone viable on a small corpus	Features are general rather than task-specific; a frozen backbone caps achievable accuracy at whatever its representation already contains
11	Z. Liu et al., 2022	A ConvNet for the 2020s	Pure convolutional network modernised with transformer-era design choices — large kernels, inverted bottlenecks, revised normalisation	Large-scale image classification, detection and segmentation benchmarks	A well-designed convolutional network matches transformers of comparable scale, so attention is not automatically the better choice	The adversarial citation for this project: it implies a transformer branch must be shown to contribute, which only an ablation can do
12	Z. Peng et al., 2021	Conformer: Local Features Coupling Global Representations for Visual Recognition	Parallel convolutional and transformer branches with a feature- coupling unit exchanging information between them at every stage	Image classification, detection and segmentation benchmarks	Local and global representations are complementary, and coupling them bidirectionally beats either branch alone	Nearest architectural neighbour to this project; couples at every stage with no learned gate, and has no temporal component or forecasting task

S. No.	Author(s) & Year	Research Paper Title	Methodology / Technique Used	Dataset / Experimental Setup	Key Findings	Limitations / Research Gap
13	C. R. Chen, Q. Fan, and R. Panda, 2021	CrossViT: Cross-Attention Multi-Scale Vision Transformer for Image Classification	Dual transformer branches at different patch scales, fused by cross-attention between their classification tokens	Image classification benchmarks	Cross-attention is an effective dual-branch fusion operator, and multi-scale tokens carry complementary information	Both branches are transformers, fusion is single-directional per exchange, and the task is single-image classification
14	J. Lu et al., 2019	ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks	Co-attentional transformer layers in which each stream's queries attend to the other stream's keys and values, in both directions	Large paired image-text corpus, transferred to vision-and-language tasks	Bidirectional cross-attention between two streams is an effective fusion mechanism — published in 2019, which is why this project does not claim it	Fuses two *modalities*; applying it across two visual encoders of the same frame is a different application, not a new mechanism
15	J. Alayrac et al., 2022	Flamingo: a Visual Language Model for Few-Shot Learning	Gated cross-attention layers inserted into a frozen language model, with learned gates controlling how much cross-modal signal flows	Large interleaved image-text corpus; few-shot evaluation	A learned gate on cross-attention is effective and stable — the gating mechanism this project uses is therefore prior art	The gate is an optimisation device reported only through downstream accuracy; its learned value is not analysed as a finding
16	T. Chen, and C. Guestrin, 2016	XGBoost	Regularised gradient-boosted decision-tree ensemble with a sparsity-aware split finder and second-order objective	Tabular machine-learning benchmarks and competition datasets	Boosted trees are extremely strong on small tabular problems — and a per-lane vehicle count sequence is a small tabular problem	No appearance information, so it cannot act when detection fails; included here as the baseline the proposed model must justify itself against
17	H. Wei et al., 2018	IntelliLight	Deep Q-learning over a phase-gated network, with a reward combining queue length, delay and phase-change penalties	Microscopic simulation calibrated against real signal and camera data from a deployed intersection	Learned control beats fixed-time control on real-world-derived demand, and the evaluation protocol matters as much as the algorithm	Single intersection, reactive state built from present measurements, and no anticipatory input
18	H. Wei et al., 2019	PressLight	Reinforcement learning with a reward derived from max-pressure theory, coordinating signals along an arterial	Microscopic simulation on synthetic and real arterial networks	Grounding the reward in a known stability result makes the learned policy coordinate without explicit communication	Pressure is computed from current queue state; the method presumes reliable per-lane vehicle counts and forecasts nothing
19	C. Chen et al., 2020	Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control	Decentralised deep reinforcement learning with pressure-based state and parameter sharing across thousands of intersections	Microscopic simulation at city scale on real road networks	Learned signal control scales to city size — which is why single- intersection control cannot be claimed as a contribution	Scale rather than sensing: still requires per-lane counts, and still acts on the present

S. No.	Author(s) & Year	Research Paper Title	Methodology / Technique Used	Dataset / Experimental Setup	Key Findings	Limitations / Research Gap
20	H. Wei et al., 2021	Recent Advances in Reinforcement Learning for Traffic Signal Control	Survey of models, state and reward designs, and evaluation practice in reinforcement-learning signal control	Survey of published work and benchmarks	Fixes what the field has settled, and documents how unevenly results are reported — which is the argument for seeded runs with confidence intervals	Surveys the state designs in use; the question of anticipatory state is left largely open, which is the opening this project uses
21	J. Schulman et al., 2017	Proximal Policy Optimization Algorithms	On-policy policy-gradient optimisation with a clipped surrogate objective, trading exact trust-region enforcement for simplicity	Continuous-control and Atari benchmark suites	A simple clipped objective matches or beats more complex trust- region methods and is stable enough to be a default	A general algorithm, not a traffic method; it contributes nothing to the domain and is used here as a standard tool
22	G. Varma et al., 2019	IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments	Dataset construction and benchmarking for detection and segmentation in unconstrained traffic, with a hierarchical label set	Ego-vehicle dashcam footage from Indian roads, including auto-rickshaw and animal classes	Unconstrained Indian road scenes are measurably harder than structured-traffic benchmarks, and the class taxonomy must be extended to describe them	Ego-vehicle viewpoint: a moving camera cannot support fixed per- lane regions that stay valid across a window, so it bootstraps detection but cannot supply the corpus
23	L. Wen et al., 2020	UA-DETRAC: A new benchmark and protocol for multi-object detection and tracking	Benchmark and evaluation protocol for multi-object detection and tracking, with detection and tracking scored jointly	Fixed elevated traffic-camera video, predominantly Chinese urban and highway traffic	Fixed elevated surveillance video is the right viewpoint for lane- level work, and joint detection-tracking evaluation is necessary for it	Lane-disciplined traffic without Indian vehicle modes, so a model trained on it does not transfer to the target stream
24	Z. Tang et al., 2019	CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification	City-scale benchmark for multi-target multi-camera vehicle tracking and re-identification, with calibrated camera geometry	Fixed multi-camera video across road intersections in the United States	Multi-camera fixed-viewpoint data at city scale is collectable and annotatable, and viewpoint calibration is what makes it usable	United States traffic, lane-disciplined, and oriented to re- identification — a task this project explicitly excludes on privacy grounds
25	S. Sugiarto et al., 2024	Base saturation flow rate (BSFR) and its effect on performance of pretimed signalized intersection with non-lane based urban heterogeneous traffic	Field measurement of base saturation flow at pre-timed signalised intersections, regressed against effective approach width, with passenger-car equivalents estimated per mode	Field survey of non-lane-based urban heterogeneous traffic at signalised intersections	Base saturation flow scales nearly linearly with approach width; passenger-car equivalents of 0.24 for a motorcycle and 0.78 for an auto-rickshaw; start lost time four to five seconds and end lost time about three seconds	Measured at specific sites, so the coefficients are not portable unaltered; it quantifies the baseline's uncertainty rather than removing it

S. No.	Author(s) & Year	Research Paper Title	Methodology / Technique Used	Dataset / Experimental Setup	Key Findings	Limitations / Research Gap
26	J. Redmon et al., 2016	You Only Look Once: Unified, Real-Time Object Detection	Detection reframed as a single regression over a grid, predicting boxes and class probabilities in one forward pass	Standard object-detection benchmarks, evaluated at real-time frame rates	Detection becomes fast enough to run per frame on a live stream, which is what makes a detector usable inside a per-step counting pipeline	Accuracy on small and crowded objects lags two-stage detectors, and an obliquely-viewed saturated junction is exactly that case
27	T. Lin et al., 2017	Focal Loss for Dense Object Detection	A reshaped cross-entropy that down-weights easy negatives so training is not dominated by background, inside a single-stage detector	Standard detection benchmarks under extreme foreground-background imbalance	Class imbalance rather than architecture was what held single- stage detectors back; correcting the loss closes most of the gap	Corrects imbalance between object and background, not between rare and common vehicle classes, which is the imbalance this project carries
28	Y. Li, X. Zhang, and D. Chen, 2018	CSRNet: Dilated Convolutional Neural Networks for Understanding the Highly Congested Scenes	Dilated convolutions behind a truncated classification backbone, regressing a continuous density map rather than counting objects	Dense crowd-counting benchmarks, including highly congested scenes	Density is recoverable where individual objects are no longer separable, and the map integrates to a count without detecting anything	Counts people rather than weighting heterogeneous vehicle classes, and yields a scene density rather than a per-lane one
29	S. R. Richter et al., 2016	Playing for Data: Ground Truth from Computer Games	Pixel-accurate labels extracted from a commercial game engine by intercepting its rendering calls, at no annotation cost	Synthetic urban driving frames, transferred to real segmentation benchmarks	Synthetic data substitutes for a large fraction of real annotation, and a small amount of real in-domain data recovers most of the remainder	The synthetic-to-real gap is real and must be measured; a model trained on synthetic alone does not match one trained in domain
30	Y. Cui et al., 2019	Class-Balanced Loss Based on Effective Number of Samples	A loss re-weighted by the effective number of samples per class rather than by raw frequency, derived from expected sample coverage	Long-tailed image classification benchmarks at controlled imbalance factors	Re-weighting by effective number beats inverse-frequency weighting as imbalance grows, which is the regime a rare-class taxonomy sits in	Assumes every class is present to be weighted, so it cannot help a class that has no test instances at all

Read down the final column, the table says something the individual rows do not. The limitations are not scattered: they concentrate on four assumptions — that sensing infrastructure exists, that a scene-level or link-level output is enough, that the present can stand in for the near future, and that traffic is lane-disciplined. Section 4 takes those four and states the gap they leave.

CHAPTER 4

Research Gap

The gap below is derived from Section 2 and Section 3 rather than asserted. Every claim about what is missing carries the citation of the work that establishes what is present, because a research gap stated without that is indistinguishable from an incomplete survey.

4.1 What has already been done

The survey establishes four settled bodies of work. Congestion can be estimated from a fixed camera's imagery, and this is an active field rather than an opening [1], [2], [3]. Spatio-temporal forecasting over an instrumented sensor network is mature and accurate [5], [4], [7], [6]. Fusing convolutional and transformer representations by bidirectional cross-attention with a learned gate is published, in some cases years ago [19], [16], [21], [23]. And reinforcement-learning signal control is heavily benchmarked, including at city scale [27], [28], [31], [32]. This project claims none of these four as its contribution.

4.2 What limitations exist in existing approaches

The accurate forecasters assume sensing that the target junctions do not have. Every method in the graph-forecasting strand consumes loop-detector or probe readings over a known road graph [5], [4], [7], [6]. At an Indian junction with a camera and no loops those inputs do not exist, so the reported accuracy is unavailable rather than merely harder to reach.

The camera-based work produces the wrong granularity. Published vision-based congestion work classifies the scene as a whole or predicts link-level flow [1], [3]. A signal controller allocates green to an approach, so a scene-level score gives it nothing to act on.

Most of it estimates the present rather than the future. Detection of current congestion [1] is a measurement. A signal decision takes effect over the next phase, so a controller needs a forecast at roughly one phase's horizon.

Evaluation settings are lane-disciplined. The fixed-camera benchmarks that have the right viewpoint carry structured traffic [35], [33], and the Indian dataset with the right traffic has a moving ego-vehicle viewpoint [34]. Neither combination matches the deployment target.

The gate is reported only through its effect. Gated fusion is evaluated by downstream accuracy [23]. What the gate learns is not itself examined, so the interpretability the mechanism could offer is left unclaimed.

Controller state is built from the present. The reinforcement-learning literature composes state from instantaneous measurements [28], [30], [31]. Whether an anticipatory input helps, and how accurate it must be before it does, is not systematically tested.

4.3 What problems remain unresolved

No published method produces a per-lane congestion forecast, at a one-phase horizon, from a single fixed camera, under non-lane-disciplined heterogeneous traffic, and feeds it to a controller. Each element exists separately; the conjunction does not.

It is not established whether appearance information beats vehicle counts for this task. The count-based alternative is cheap and strong [25], and the comparison is rarely made on verified labels with whole cameras held out.

It is not known whether a fusion gate learns anything that corresponds to a physical property of the scene, because the works that use gates do not report their values [23].

The sensitivity of a learned controller to forecast quality is unmeasured. Without it, a positive control result cannot be attributed to anticipation rather than to the controller itself.

There is no public corpus of continuous fixed-camera video of heterogeneous Indian junction traffic with per-lane labels. The datasets that exist have either the viewpoint or the traffic, not both [34], [35], [33].

4.4 What has not been adequately addressed

Congestion defined on occupancy rather than count. A count treats a bus and a two-wheeler alike. Under heterogeneous traffic the quantity a controller needs is occupancy weighted by passenger-car units, and the published equivalences differ enough between conventions to change the answer [46], [50], [51].

Per-camera threshold calibration. A congestion threshold that is correct at one viewpoint is arbitrary at another, yet thresholds are commonly fixed globally.

Camera-level generalisation. Splitting overlapping windows at random lets a model recognise a camera it has already seen and report that as generalisation. Holding out whole cameras is the only split that measures what deployment requires.

Labels that do not derive from the baseline's own inputs. If congestion labels are derived from detector counts and the baseline also consumes detector counts, the comparison is circular. A human-verified test split is what breaks it.

Privacy as a design constraint rather than a policy. Work in this area generally assumes footage can be moved and stored. Keeping raw frames on the device and transmitting only derived counts is a constraint on the architecture, not a statement added afterwards.

4.5 How the proposed project addresses the identified gap

Claim the conjunction, not the components. The contribution is stated as per-lane congestion forecasting from a single fixed camera under non-lane-disciplined heterogeneous traffic, coupled to a controller. Every component mechanism is cited as prior art rather than presented as new.

Build the corpus the gap implies. Curate continuous fixed-camera footage of heterogeneous traffic from permissively-licensed sources, place lane regions per camera by hand, and label per-lane congestion from passenger-car-unit weighted occupancy with per-camera thresholds.

Make the comparison that decides the question. Evaluate against count-only forecasters including a boosted-tree model [25] and a linear probe, on a human-verified test split, with whole cameras held out — and report the result whichever way it goes.

Treat the gate as a measurement. Log the gate value per window and analyse it against scene density, reporting a gate that collapses or tracks nothing as the negative result it is.

Answer the control question with a curve, not a comparison. Sweep forecast quality from none through a noise-calibrated surrogate to an oracle, so the result says how good the forecast must be before anticipation pays, and compare against the best-tuned fixed-time baseline [49] rather than an arbitrary one.

Design for the privacy constraint. Run detection at the edge and transmit only derived counts and predictions, so that no raw frame is written to disk or crosses the network at any point in the pipeline.

4.6 The gap, stated once

Stated plainly, the gap this project occupies is not a missing mechanism. The mechanisms exist and are cited. It is the absence of a method that works from the sensing an Indian junction actually has, produces the quantity a signal controller actually consumes, is

evaluated under the traffic that actually arrives, and is honest about which of its parts earn their cost.

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